

HORACIO MORENO MONTAÑES

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EDUCATION

University of Michigan, Ann Arbor

Ph.D. in Applied and Interdisciplinary Mathematics and Scientific Computing

August 2024 – Present

Advisor: Robert Krasny

University of Michigan, Ann Arbor

B.S. in Mathematical Sciences, B.S. in Interdisciplinary Physics (GPA: 3.928)

May 2024

Relevant Graduate Coursework

- MATH 571: Num. Linear Algebra
- MATH 572: Num. Methods for Differential Equations
- MATH 526: Stochastic Processes
- MATH 654: Intro to Fluid Mechanics
- NERS 571 & 572: Intermediate Plasma Physics I & II
- SPACE 597: Space Plasma Physics
- MECHENG 523: Computational Fluid Dynamics
- EECS 471: GPU Programming

RESEARCH INTERESTS

My research focuses on the development and scaling of numerical methods for physical systems.

- **Numerical Methods:** Particle methods with adaptive tree-based grids for plasma and fluid dynamics.
- **Scientific Computing:** High-performance parallel algorithms and GPU acceleration for multi-scale phenomena.
- **Data-Driven Modeling:** Development of empirical models for complex physical systems.

PUBLICATIONS

H. Moreno Montañes and E. Landi, “Empirical Modeling of the Fast Solar Wind,” *The Astrophysical Journal (ApJ)*, **Accepted Dec. 2025**.

RESEARCH EXPERIENCE

REU at the University of Michigan

Modeling 1D Cold Electrostatic Plasma (Advisor: Robert Krasny)

May 2023 – Present

- Developed a semi-Lagrangian particle method in Python using Vlasov-Poisson equations.
- Implemented electric field regularization and particle insertion to maintain plasma definition.
- Analyzed method convergence regarding timestep size and spatial discretization.
- Investigated applications for dark matter perturbations at cosmological scales.

Math 462: Introduction to Mathematical Modeling

Coursework Research Projects

Jan 2023 – April 2023

- Conducted discrete modeling of dune-obstacle interactions and stadium wave propagation.
- Built simulations in Matlab and Python to assess accuracy against real-world systems.

PRESENTATIONS

Conference Presentations

- *Computation of Potential Magnetic Fields in the Solar Corona with Integral Equation Methods*, MIPSE Graduate Symposium, Ann Arbor, MI Dec 2025
- *Modeling 1D Cold Electrostatic Plasma with a Lagrangian Particle Method*, SACNAS NDiSTEM Conference, Portland, OR Oct 2023
- *Empirical Modeling of Solar Wind*, Undergraduate Research Symposium, Ann Arbor, MI April 2021

Seminars & Invited Talks

- *Barycentric Lagrange Tree-Traversal Fast Summation Algorithms*, Student AIM Seminar, University of Michigan Nov 2025

TEACHING AND MENTORING

University of Michigan

- **Instructor** MATH 389: Explorations in Mathematics Research Winter 2026
- **Graduate Student Instructor** MATH 115: Calculus II Winter 2025, Fall 2025
- **Graduate Student Instructor** MATH 116: Calculus I Fall 2024
- **Undergraduate Course Assistant**

MATH 389: Explorations in Mathematics Research Winter 2023, Winter 2024
MATH 286: Honors Calculus III Fall 2023

Directed Reading Program (DRP)

Graduate Mentor for Undergraduate Students

May 2025 – Present
University of Michigan

- Nonlinear Dynamics and Chaotic Systems Winter 2026
- Introductory Probability Fall 2025
- Numerical Methods for ODEs Summer 2025

TECHNICAL SKILLS

Languages	Python, MATLAB, C++, Mathematica
Libraries	NumPy, SciPy, CUDA, MPI, Kokkos
Computing Utilities	Slurm, Git, Bash

HONORS AND AWARDS

MICDE Graduate Fellowship , Michigan Institute for Computational Discovery & Engineering	2024
Wilfred Kaplan Award in Applied Mathematics , University of Michigan	2024
SACNAS NDiSTEM Conference Travel Scholarship	2023
James B. Angell Scholar , University of Michigan	2022, 2023
Ferrante Scholarship Fund Recipient (Four-year merit award)	2020 – 2024